OctaSence

The Intelligence Layer for

Critical Infrastructure Safety

Predict failures before they happen — across mines, dams, tunnels & civil infrastructure

$10B+
Market Size

€2M
Seed Round

B2B SaaS
Model

AI + IoT Platform
for Infrastructure
Intelligence

THE PROBLEM

Infrastructure Failures Are Sudden, Deadly, Expensive — And Mostly Preventable

Where Failures Occur

⛏️ Mining — pillar/roof collapses, slope failures

🌊 Dams — seepage → liquefaction → breach

🚇 Tunnels — ground settlement, water ingress

🌉 Civil/Rail — bridge fatigue, embankment washouts

🌿 Environmental — landslides, groundwater

What's Actually Broken

❌ Periodic inspections (weekly/monthly)

❌ Isolated sensors with no intelligence

❌ Threshold alarms — no prediction

❌ Manual interpretation under stress

❌ Offline failure in remote sites

Why It's Systemic

- Infrastructure degrades continuously; monitoring is discrete
- Data exists, but no unified risk intelligence
- Compliance is retrospective, not preventive
- Human judgment doesn't scale with asset growth

➡ *Risk becomes visible only when failure is already in motion. The real failure is informational.*

WHY NOW

Four Converging Forces Create an Urgent, Winnable Opportunity

01 Infrastructure Scale

More at Risk

- Global critical infra market growing with trillions in active assets.
- India's €1.037 trillion NIP alone adds massive exposure.
- More infrastructure = more failure points unless intelligence scales.

02 Climate Volatility

Faster, Non-Linear Risk

- Extreme rainfall, rising groundwater & seasonal shocks accelerate failures.
- Climate-driven events evolve faster than inspection cycles.
- Historical data can't keep up with real-world change.

03 Regulatory Pressure

Proof of Prevention Required

- Regulators, insurers & ESG frameworks demand continuous due diligence.
- Liability & insurance premiums depend on demonstrable prevention.
- 'Inspected last month' is no longer a defensible position.

04 Technology Inflection

Deployable Now

- Low-cost rugged sensors, Edge AI & digital twins crossed feasibility threshold.
- Continuous predictive monitoring is now technically viable & economically scalable.
- What was experimental a decade ago is deployable today.

THE SOLUTION

OctaSence = Real-Time Infrastructure Intelligence

A unified AI + IoT platform that turns infrastructure from being monitored to being intelligent.

🧠 AI Agents

Autonomous risk reasoning, anomaly detection & early-failure prediction

🌐 Multimodal IoT

Geotechnical • Structural • Environmental • Vision — fused in real time

OctaSence

⚡ Edge Computing

Local AI inference for remote & offline mines, dams, tunnels

📐 Digital Twins

Live 2D/3D views of stress, deformation, seepage & risk evolution

🧩 Geotechnical Models

Physics + domain models embedded into AI predictions

HOW IT WORKS

Six Layers of Intelligence — From Ground Truth to Compliance

1 Multimodal Sensing

Ground Truth Layer

Strain, stress, vibration, tilt, seepage, groundwater, vision inputs — continuously fused

2 Edge Intelligence

Reality Layer

Offline-first operation with store-and-forward — safe even when network fails

3 AI & Geotech Reasoning

Intelligence Core

LSTM/GBM models predict failure windows (6–72h); physics + data, not black-box ML

4 Digital Twin & Risk Viz

Decision Layer

Live 2D/3D heatmaps, time-lapse replay, risk propagation — spatial decisions under pressure

5 Alerts & Preventive Action

Response Layer

Multi-channel: WhatsApp, SMS, Email, API. Critical alert latency <5–10 seconds

6 Continuous Learning

Trust & Compliance Layer

DGMS/ISO/BIS-ready reports, full audit trail, model recalibration & sensor drift detection

CORE PRODUCTS

Four Modules. One Integrated Platform.

MONITORING

- Continuous multimodal sensing (strain, tilt, vibration, seepage, vision)
- Edge-processed data with quality & health checks
- Real-time site visibility with zero blind windows

→ Zero blind spots

PREDICTION

- Physics-aware ML predicts failure windows (6–72h)
- Pillar, slope, bench & dam risk forecasting
- Explainable risk scores + root-cause hints

→ Hours to days of lead time

DIGITAL TWIN

- Live 2D/3D twins with risk heatmaps & simulations
- Time-lapse replay, what-if scenario analysis
- API integration for GIS, ERP, SCADA

→ Spatially precise decisions

COMPLIANCE & AGENTS

- Automated DGMS / ISO / BIS reports + audit trails
- AI agents: daily safety, audits, RCA, data quality
- Multi-channel dispatch: WhatsApp, SMS, Email, API

→ Continuous due diligence

COMPETITIVE ADVANTAGE

Why OctaSence Wins — And Why It's Hard to Replicate

Built for Prediction, Not Monitoring

- Unified Intelligence Stack — AI Agents + Multimodal IoT + Geotechnical Models + Digital Twins
- Predictive by Design — failure forecasts hours to days ahead, not threshold alarms
- Geotech-First AI — physics-aware models trained on Indian + global geology

This is not a dashboard — it's an intelligence layer.

Fits Reality, Scales Enterprise-Wide

- Edge & Offline-First — local inference for deep mines & remote dams
- Sensor-Agnostic & SCADA-Friendly — integrates with existing instrumentation
- Multi-Asset, Multi-Site Ready — scales from 1 structure to 100+ sites seamlessly

Designed for real sites, not lab conditions.

Built for Regulators, Insurers & Boards

- Compliance-Ready by Default — automated DGMS / ISO / BIS reports + full audit trails
- Explainable & Tamper-Proof AI — inference logs, traceability, secure pipelines
- Proven Operational Metrics — critical alerts <5–10s | delivery >95% | 99% uptime target

Proves prevention — not excuses after failure.

MARKET OPPORTUNITY

Large TAM. Focused SAM. Credible SOM.

TAM

$10B+ 5-Year | India + EM

Mining, dams, civil infra, railways, tunnels, environmental. All infrastructure safety, monitoring & predictive intelligence spend.

SAM

$1–2B Near-Term Focus

High-risk mines, tailings dams, metro tunnels, critical bridges, compliance-driven PSU / EPC projects.

SOM

$150–300M 3–5 Year Capture

Pilot-led land-and-expand. Multi-site SaaS + hardware + services. Long-term government & PSU rollouts.

COMPETITIVE LANDSCAPE

Point Solutions Exist. Unified Infrastructure Intelligence Does Not.

Capability	OctaSence	Worldsensing	IDS GeoRadar	Trimble	Froq.io
Core Focus	**AI + IoT + SHM**	Geotech IoT	LiDAR / GPR	Survey & Monitoring	IoT + AI
Domains	**Mining + Civil**	Mining	Mining + Infra	Infra	Bridges/Tunnels
Predictive AI	🟢 **Advanced**	⚪ Basic	🔴 None	⚪ Basic	🟡 Basic
Multimodal Fusion	🟢 **Full Fusion**	🟡 IoT Only	🔴 Radar Only	🟡 Survey	🟡 IoT + AI
Edge / Offline AI	✅ **Yes**	❌ No	❌ No	❌ No	⚪ Limited
Digital Twin	✅ **Integrated**	❌ No	❌ No	❌ No	⚪ Basic
Compliance Auto	✅ **DGMS/CWC/PWD**	❌ No	❌ No	❌ No	⚪ Partial
Deploy Speed	🚀 **Very Fast**	Medium	Slow	Medium	🚀 Fast

OctaSence is the only platform combining full-stack prediction + multimodal fusion + edge AI + compliance automation.

BUSINESS MODEL

Scalable, Recurring Revenue with High Retention

1

SaaS Subscription

Core Revenue

- Annual/multi-year subscription
- Per site/asset/sensor tier
- AI dashboard, risk scoring, alerts, digital twins
- Software margins improve with every new site

ARR Engine

2

Hardware & Edge Devices

Sale or Lease

- Rugged sensors + edge gateways
- Sale (CAPEX) or lease (OPEX) model
- Bundled with SaaS for faster adoption
- Low upfront friction → predictable expansion

Low Friction

3

Deployment & Services

High-Margin Services

- Site assessment & sensor layout design
- Geotechnical model calibration
- SCADA / ERP / GIS integration
- Accelerates go-live, improves prediction accuracy

Sticky

4

Enterprise Add-Ons

Upsell & Expansion

- Advanced analytics & prediction packs
- Compliance automation (DGMS / ISO / BIS)
- AI agents: reports, audits, RCA
- Government & PSU long-term contracts

Compounds ACV

FY26 Revenue: €360K · FY27 Revenue: €1.53M · FY28 Revenue: €3.96M · Gross Margin FY28: 91.5% · EBITDA+ by Month 22

TRACTION & PILOTS

Validated Problem. Proven Technology. Execution Underway.

Market Validation

The Problem Is Real

100+ Industry Professionals Engaged

Mining • Dams • Civil Infra • Rail • Safety

Reactive monitoring confirmed as #1 pain

Rising compliance pressure — operators & regulators

No predictive early-warning available today

Confirmed across operators, EPCs, consultants & regulators

Product Validation

The Technology Works

AI Models Validated on Real Patterns

✔ Sensor anomaly detection

✔ Trend deviation identification

✔ Risk escalation signals

✔ Edge + offline workflows proven

Core AI + edge + geotechnical thesis is proven.

Pilots & Execution

From Intent to Deployment

5 Active Pilot Customers Onboarded

⛏️ Mining operators

🏗️ Infrastructure / EPC-linked projects

📍 Pilot scope defined per site

🔧 Sensor & edge installation in progress

Clear transition from intent → deployment.

🏛️ Regulatory access via Ministry of Mines advisors (NIRM/DGMS) — govt demand for continuous monitoring validated

PRODUCT & EXECUTION ROADMAP

From MVP → Market Standard → Global Infrastructure Intelligence

2026

Q1–Q3

MVP & Field Pilots

- MVP Platform: dashboards, AI anomaly detection, edge ingestion
- Predictive alerts (6–72h window), WhatsApp/SMS/Email
- 5–10 live pilot sites: mines, dams, tunnels, civil infra
- DGMS-aligned compliance v1 reports

→ Product-market validation

2027

Commercial V1

Early Scale

- InfraEdge V1: multi-site dashboards, digital twins 2D/3D
- Multimodal AI fusion + mobile app for safety officers
- AI Agents: daily safety, audit, RCA agents
- RBAC, multi-tenancy, SCADA/ERP APIs

→ 30+ paying customers

2028

Category Leadership

India + EM Domination

- Digital Twins v2: simulation & what-if scenarios
- OctaSence in-house sensor stack (lower COGS)
- Remaining Useful Life (RUL) prediction module
- 100+ paying customers, PSU + govt adoption

→ EBITDA positive, Series A ready

2029–30

Global Expansion

World's Infra Intelligence Layer

- Expand to APAC, LATAM, Africa
- National-scale govt dashboards
- Autonomous emergency response AI agents
- High ARR, strong regulatory moat

→ Category-defining platform

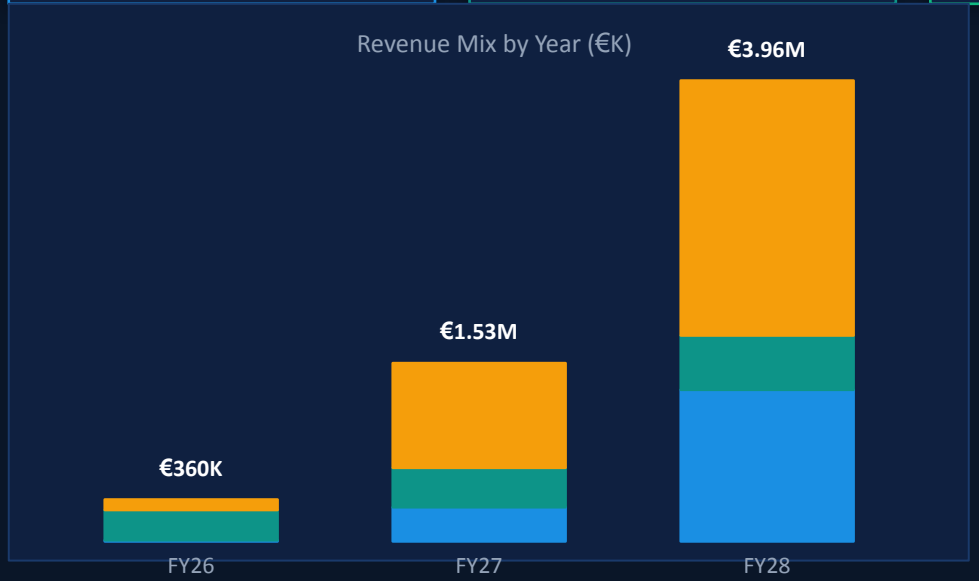
3-Year Financial Summary — All Figures in EUR

FY26 Revenue
€360K
Hardware & Services

FY27 Revenue
€1.53M
SaaS + H/W + Services

FY28 Revenue
€3.96M
SaaS-led, EBITDA+

3-Year Total
€5.86M
ARR €6.58M by FY28



Metric	FY26	FY27	FY28
Gross Margin %	59.3%	80.3%	91.5%
EBITDA	–€595K	–€376K	+€1.59M
Cash Balance	€1.40M	€1.03M	€2.62M
ARR End of FY	€1.02M	€2.73M	€6.58M

SaaS free M1–M8 (pilot programme) · SaaS pay-per-use from M9 · EBITDA breakeven Month 22–24

THE TEAM

Experienced Founders. Deep Domain. Global Reach.

Shivaraj Choutagi

CEO

IIM-K | Ex-Cofounder Picake | Member MEAI & ICMM

Harsh Vardhan

CTO

IIT-Delhi | KPMG | X-Housing.com | X-IBM | 2× Founder

Wolfgang Staufer

CBO

TU Wien | Ex-CEO & VP SinAura | European Sales

Vasiliy Bezlyudnyy

CPOO

TalTech | Product Owner FluxHub Estonia

SCIENTIFIC & ADVISORY BOARD

Dr. Sripad R. Naik

Scientific & Strategic Advisor

Director, NIRM | Ministry of Mines, Govt. of India

Dr. Aishwarya T

Geotechnical Consultant & SME

IIT B | PhD Geotech | Member IGS & ISSMGE

Dr. P. C. Jha

Scientific Advisor

Former Scientist, NIRM, Govt. of India

THE ASK

Finance Sheet Link

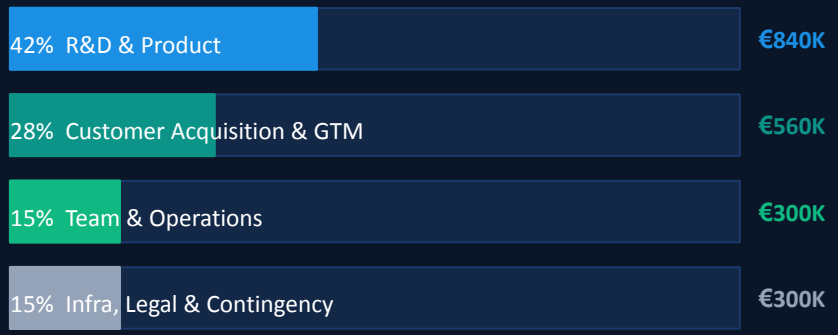
Capital to Convert Pilots → Customers → Category Leadership

€2M

Seed / Pre-Seed Round

24 Months Runway · Feb 2026 – Jan 2028

USE OF FUNDS



24-MONTH OUTCOMES

🚀 Convert 5 pilots → 45+ paying enterprise customers

🧠 Deliver InfraEdge V1 — AI risk models, digital twins, predictive alerting

🏛️ Prepare platform for PSU, government & regulator-led rollouts

💰 EBITDA breakeven by Month 22 (FY28 Q4)

📡 Launch in-house sensor + edge stack — improved margins & defensibility

Beyond Capital:

🤝 Pilot & Scale Partners (mines, EPCs, infra owners) 🌍 Govt & PSU Alliances 🧠 Domain Advisors (geotech, safety, large infra ops)

Octa Sence

The intelligence layer that prevents infrastructure failures before they happen.

Prediction is not a feature. It is the foundation.

Shivaraj Choutagi — CEO

IIM-K | Member MEAI & ICMM

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